

Spectrum width

Y14 (2014) — English translation

To assess the degree of monochromaticity of the source emitting in the vicinity of the wave line $\lambda = 500$ nm, a Young interference scheme was used (the distance between the slits in the opaque screen $d = 1$ cm). The observation plane was moved into the Fraunhofer zone. Only bands whose intensity was at least 4 % the intensity of the zero band were observed visually. It turned out that when the width of the slits changes from $b_1 = 0.5$ mm to $b_2 = 0.1$ mm, the number of observed bands increases from $N_1 = 40$ to $N_2 = 80$.

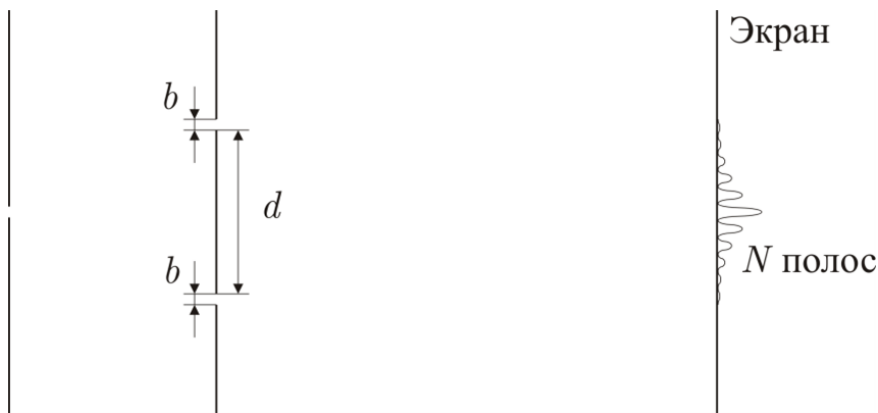


Figure 1: Figure 1.

A1	Estimate the width of the emission spectrum $\Delta\lambda/\lambda$.
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3.00

Source: <https://pho.rs/p/3993>